Dhruval Javia

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Education

Stanford University Expected June 2026
M.S. in Mechanical Engineering (Specialization: Energy and Transport Sciences), GPA 3.73/4.00 *Stanford, CA*

- **Relevant Coursework:** Heat Conduction, Convective Heat Transfer, Physical Gas Dynamics, Gas-phase Chemical Kinetics, Optical Diagnostics and Spectroscopy Lab, Batteries & Fuel Cells, FEA, Numerical Methods, Projection-based Model Order Reduction

Indian Institute of Technology, Bhubaneswar Aug 2018 – April 2022
B.Tech. in Mechanical Engineering *Bhubaneswar, Odisha*

Work Experience

Thermax Ltd. Aug 2022 – July 2024
Graduate Trainee – Advent Rotational Program *Pune, Maharashtra*

- Mapped manufacturing processes for boilers and heaters on the factory floor, identified bottlenecks, and proposed automation improvements; prepared detailed RLA and RCA reports based on site evaluations.
- Led a cross-functional team to optimize and standardize biomass-fired thermal oil heater design, achieving a 7.4% weight reduction and 10.2% smaller footprint.
- Performed pressure part sizing, thermal & pressure drop analysis and weight estimation to optimize techno-commercial offering of the heater.
- Developed thermal models of the APH, Economizer and MPA furnace in Excel with VBA automation to iteratively and rapidly design the heater, and validated it with existing data. The model provided inlet/outlet temperature values within 2%, 6% and 10% error in APH, Economizer and MPA furnace respectively.

Summer Research Internship May 2021 – July 2021
IIT Madras, Project: Macro-scale Design Aspects of EV Battery Pack *Online*

- Performed battery pack design calculations by developing a system of equations in Excel and implementing it in a MATLAB app. The app sliders were used to analyze the effect of battery pack design parameters on EV performance.
- Developed MATLAB script to simulate single-cell current profiles by integrating vehicle dynamics equations with standardized velocity profiles.

Academic Projects

Efficient Battery Thermal Management Systems For EV | *B.Tech. Thesis Project* July 2021 – April 2022

- Performed degradation studies using COMSOL Multiphysics via coupled 1-D P2D electrochemical and 2-D CFD thermal models of an air-cooled battery pack integrated with a capacity fade model.
- Optimized cell arrangement in air-cooled pack for minimum peak pack temperature and cell-to-cell temperature difference considering cell degradation due to SEI formation.
- Identified 26650 cells and tapered arrangements as the ideal combination to minimize volumetric heat generation and peak pack temperatures.
- Published the work in the “Energy Storage” journal having an impact factor of 3.2.

Aerocase | *Individual Course Project, ME 203: Design and Manufacturing, Stanford University* April 2025 – June 2025

- Designed and fabricated a compact, innovative, and functional instrument case ensuring the safe storage of Aerodrum instruments, with a time constraint of 10 weeks and a budget constraint of $100.
- Fabricated the case using TIG welded 0.04” thick Al 5052 aluminum sheet, A356 cast aluminum, and pine wood.

Gas Dynamics Simulator | *Individual Personal Summer Project* June 2025 – Aug 2025

- Built a MATLAB program from scratch using first-principles, to simulate particle dynamics and association chemical reactions in an ideal gas, implementing hash grid optimization to reduce no. of collision checks by a factor of 200.
- Developed an interactive MATLAB app with numeric and GUI-based input to improve user experience, providing time-series plots and dynamic animations as post-processing outputs for analysis.

Technical Skills

Programming Languages: MATLAB, Python
Simulation: ANSYS, COMSOL, Firedrake
CAD: SolidWorks, Fusion 360, OnShape
Manufacturing: Sheet Metal Working, TIG Welding, Sand Casting, Woodworking, 3D Printing

Leadership and Achievements

Student Placement Coordinator (2022): Coordinated recruitment processes with industry and institute stakeholders.
Team Lead, e-Yantra Robotics Competition (2019-20): Collaborated with a 4-member team in the IIT Bombay robotics competition.
5th Rank, GrabCAD Shop Challenge: Ranked in the top 6% of participants globally for the design of a 3D-printable mechanical hand blender.